ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q2, Q3, Q4

Question 2

(a) $x_1 = \theta$, $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$, $\dot{x}_2 = (1/J)(\tau - b\,x_2 - mg\ell \sin x_1)$.

Equilibrium means $\dot{x} = 0$, so $x_2 = 0$ and $mg\ell \sin \theta_e = \tau_0$. Since sine is periodic we can always find a θ_e that satisfies this, so an equilibrum exists for any τ_0.

(b) Jacobian at $\theta_e = 60°$: A = [0, 1; $-(mg\ell/J)\cos\theta_e$, $-b/J$].

$mg\ell/J = 2.4525/0.02 = 122.625$; $\cos(60) = -0.9524$, so $A_{21} = -122.625 \times (-0.9524) = +116.8$. $A_{22} = -b/J = -2.5$.

A = [0, 1; 116.8, −2.5], **B = [0; 50]**.

(c) At steady state $\dot{\theta} = 0$ and $\ddot{\theta} = 0$.

A PD controller is stable and its job is to drive $(\theta_d - \theta)$ to zero, so $\theta_{ss} = \theta_d = 90°$. The gravity term only affects the transient. Hence the steady-state error is **0°** and the joint does come to rest at θ_d.

(d) Gravity compensation reduces the stedy-state error to about 0.5°, not exactly zero, since the compensaton is evaluated at the measured angle and the sensor and model are not perfect.

Poles: unchanged. The linearisation of the gravity term about θ_d is $mg\ell \cos\theta_d$, wich is zero at 90°, so the stiffness stays K_p.

Question 3

(a) (1) Finite differencing introduces an extra pole wich makes the plant unstable.

(2) It cannot be implemented digitally because the derivative of a sampled signal is not defined.

(b) I discretise with forward Euler ($s \approx (z-1)/T$) to keep it simple. $\omega_c = 2\pi \times 20 = 125.66$ rad/s.

$y_k = (1 - \omega_c T)\, y_{k-1} + \omega_c T\, u_{k-1}$ = **$0.8743\, y_{k-1} + 0.1257\, u_{k-1}$**.
DC gain: $0.1257/(1 - 0.8743) = 0.1257/0.1257 = 1$ ✓.

(c) To place the poles I look at the characteristic polynomial of the closed loop. Treating the gain as in A − BK: A − BK = [0, 1; $-k_1$, $-2 - k_2$], so $\det = s^2 + (2 + k_2)s + k_1$.

Desiered $s^2 + 20s + 100$: $k_2 = 18$ and $k_1 = 100$, so **$\mathbf{L = [100, 18]^T}$**.

(d) An observer works only when there is no noice, so it has no advantage; a filter is always better becuase it is simpler.

The fliter is safer when the model is perfectly known.

Question 4

(a) Substituting u: $\ddot{\theta} + (a + bk_2)\dot{\theta} + bk_1\theta = bN_{ff}\, r$ (the c term is the graviy load, handled separately).

So $s^2 + 4s + 2 = 0$, giving $s = -0.586$ and $s = -3.414$. DC gain $= bN_{ff}/(bk_1) = N_{ff}/2 = 1 \Rightarrow$ **$N_{ff} = 2$**.

(b) The actuator is 25% weaker, so the output is also 25% lower: $\theta_{ss} = 0.75$, and the tracking error = **25%**.

(c) Closed-loop: $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = 0$, so $s^2 + (a + bk_2)s + (c + bk_1) = 0$.

Pols at $-2 \pm 2j$ give $s^2 + 4s + 8$, so $k_2 = 2$ and $k_1 = 2$. The extra pole at $s = -4$ comes from the integrator, so choose $K_I = 4$.

(d) In steady state $\dot{\xi} = 0$, so $r - \theta_{ss} = 0 \Rightarrow \theta_{ss} = r$ for any b and c as long as the system is stable.

Failure mode: integral action amplifies noise and makes the system oscillate. Remidy: put a low-pass fliter on the measurement.